

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458281.90189 +/- 0.00083 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.179 +/- 0.04
Transit depth (Rp/Rs)^2	3.22 +/- 1.43 [%]
Orbital Inclination (inc)	83.96 +/- 1.77
Airmass coefficient 1 (a1)	1.997 +/- 0.097
Airmass coefficient 2 (a2)	-0.62 +/- 0.16
Scatter in the residuals of the lightcurve fit is	4.86 %
Best Comparison Star	None
Optimal Aperture	3.65
Optimal Annulus	11.97
Transit Duration (day)	0.085 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.179 ± 0.04 vs 0.13031 (deviation 1.0σ)
Duration fit vs published	0.085 d vs 0.0954 d (deviation 0.78σ)
Mid-time O–C	2.0 min
Depth SNR	3.7
Residual scatter	4.86 %

Light curve

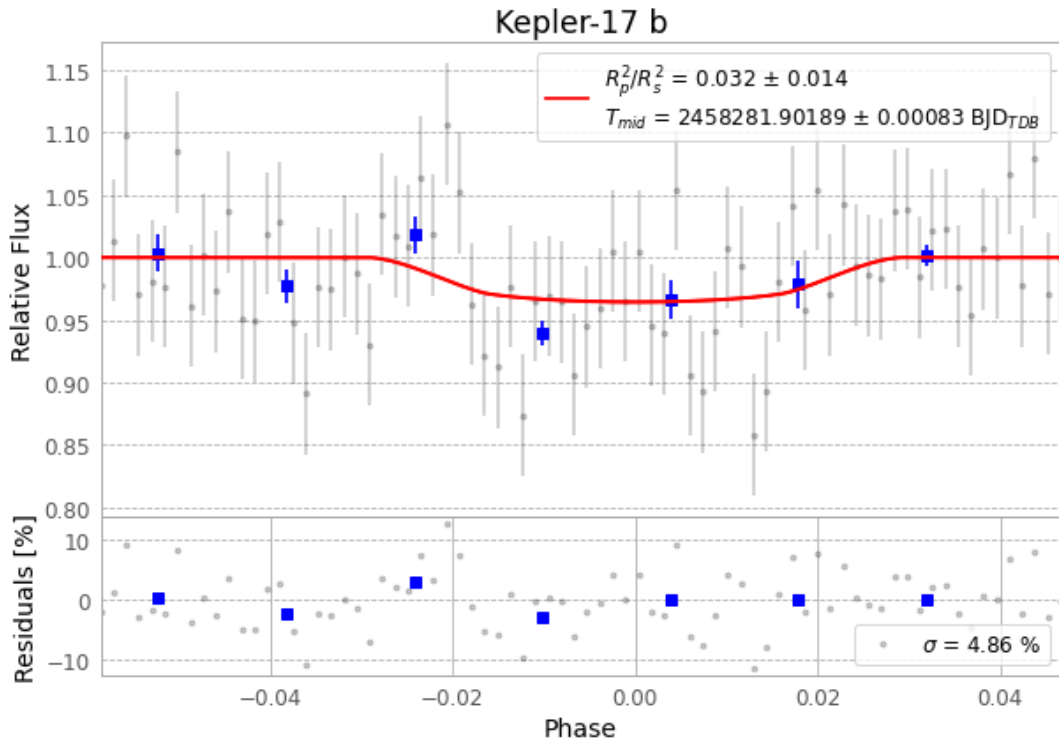


Figure 1 — FinalLightCurve_Kepler-17 b_2018-06-12.png (SHA-256 fa7e81a2719485ab...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [421, 251], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c6a93ffe714bb8a7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

76 FITS frames (50 MB), KEPLER-17180612073012.FITS → KEPLER-17180612111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-KEPLER-17-180612.log (ff95fc197392...), FinalLightCurve_Kepler-17 b_2018-06-12.png (fa7e81a27194...), FinalLightCurve_Kepler-17 b_2018-06-12.csv (65dbd06275c8...)

Compute resources

Wall time 123.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 01:44Z.